Problem Statement: Chemical Accuracy at Scale for Density Functional Theory

Prompt: Develop methods that bring the accuracy of high-level quantum chemistry (≈1 kcal/mol, "chemical accuracy") to the properties that decide real materials choices, at a cost low enough to apply to thousands of systems of thousands of atoms. Density functional theory (DFT) is exact in principle for the ground-state energy and density [1, 2], but all unknown physics is carried by the exchange–correlation (XC) functional, which must be approximated; today's approximations are off by 2–10 kcal/mol and fail qualitatively for strong correlation and charge transfer [3, 4]. For quantities that are not ground-state properties, above all the band gap, the Kohn–Sham eigenvalue gap differs from the fundamental gap by the derivative discontinuity even for the exact functional [5], so the theory itself must be extended [6, 7]. And standard Kohn–Sham DFT scales as $O(N^3)$ with system size, hybrids worse, capping routine calculations near 10^3 atoms although locality arguments say linear scaling is possible [8, 9].

Given a set of molecules or materials (composition, geometry, charge and spin state), predict their ground-state energetics, electronic structure and response properties to within stated tolerances of CCSD(T), quantum Monte Carlo or experimental references, with a computational cost that scales linearly with the number of atoms, and with the accuracy traceable to physics (exact constraints, derived forms, or machine-learned constructions that satisfy exact conditions by construction) rather than to fitting on the test data.

The method need not be a single functional or a single algorithm. New exact conditions on the functional, new semilocal, nonlocal or orbital-dependent forms, machine-learned functionals, linear-scaling solvers, embedding schemes and extensions of the theory (generalized Kohn–Sham, time-dependent, relativistic, superconducting DFT) are all acceptable. The goal is a shared, physically grounded description whose accuracy transfers to systems and property classes it was not developed on, not a collection of unrelated fits.

The method should predict the following systems. The outputs below are proposed challenge requirements. References provide the theoretical basis and existing benchmarks for each class; they do not establish that the requested accuracy at the requested cost has already been demonstrated.

Calibration Systems (public development set)

System	Required predictions
First-row small molecules (H_2, LiH, CH_4, NH_3, H_2O, HF, N_2, CO, F_2, C_2H_2, HCN, H_2CO, OH, $^3CH_2$) and atoms	Atomization energies to ≤ 1 kcal/mol against CCSD(T)/CBS at fixed geometries; vertical ionization potentials of B, F and HF. Baselines on this set: PBE 7.3 / 5.2, r²SCAN 2.6 / 6.3, B3LYP 1.7 / 6.4, ωB97M-V 1.3 / 1.3 kcal/mol (atomization / IP mean absolute error).
Exact and near-exact densities (H, He, Li, Be, hydrogenic ions, Hooke's atom, uniform gas)	Satisfaction of known exact conditions [10–13]: uniform-gas limit, spin and coordinate scaling, gradient expansion, Lieb–Oxford bound, one-electron self-interaction freedom, piecewise linearity in fractional charge and the flat-plane condition. Each violated condition must be declared.
Bulk crystalline silicon and a periodic water box, 64 to 10^4 atoms	Total energy and forces within ≤ 1 meV/atom and ≤ 10 meV/Å of the conventional $O(N^3)$ calculation with the same functional; measured wall-time and memory scaling exponent.

What makes the method general?

Use shared physical descriptions: exact constraints and appropriate norms for the functional, locality (nearsightedness) for the solver, and a declared set of ingredients (density, gradient, kinetic-energy density, exact exchange, nonlocal correlation). Material-specific or class-specific parameters are allowed only when they are derived from exact conditions or from data the submitter generates independently of the benchmark; their origin must be

stated before prediction. Do not fit a new parameter set to each property class, and do not tune any parameter to the development or hidden sets. Selection among a small number of physically motivated candidate forms using development-set scores is permitted if every candidate and its score is disclosed.

Recover established limits where their assumptions hold (the uniform gas, the slowly varying gas, the one-electron limit), but permit different mechanisms where the physics differs: charge-transfer and stretched-bond systems are governed by the derivative discontinuity and fractional-charge behaviour rather than by semilocal corrections [5, 14]; band gaps require a generalized Kohn–Sham or many-body treatment [6]; metals and small-gap systems defeat locality-based truncation at zero temperature and need finite-temperature or Fermi-operator formulations [9].

Evidence and validation

The method applied to the calibration systems must be applicable, without re-parametrization, to the systems below. Final scoring uses hidden sets of different molecules and property classes; a submission is ranked on the hidden property classes absent from calibration and must not be worse than the best existing functional on any class by more than a stated tolerance.

Validation system	**Required predictions**
Unseen first-row molecules (e.g. C_2H_4, CO_2, H_2O_2, O_2, NO, CH_3, NH_2, CH_3OH, HNO, LiF, BH_3, BF) and atomic/molecular ionization potentials	Atomization energies and IPs to ≤ 1 kcal/mol against CCSD(T)/CBS, demonstrating that calibration-set accuracy transfers to new molecules of the same classes.
Second-row molecules (SiH_4, PH_3, H_2S, HCl, Cl_2, SO_2, CS), reaction barrier heights ($H + H_2$), noncovalent complexes (water dimer), proton affinities (NH_3, H_2O)	The same tolerances on property classes and elements absent from calibration; self-interaction-sensitive barriers and dispersion-bound complexes must both be captured. Existing reference: ωB97M-V achieves 2.7 / 0.7 / 0.2 / 2.1 kcal/mol on these four classes; B3LYP 1.9 / 5.3 / 0.6 / 0.6.
Main-group and transition-metal thermochemistry databases (GMTKN55 [15], transition-metal reaction sets)	Weighted mean absolute error below the best functional of the same rung of Jacob's ladder, with no subset worse than that functional by more than the tolerance.
Solids: lattice constants, cohesive energies, formation enthalpies and fundamental band gaps of a fixed set of semiconductors, insulators, metals and transition-metal oxides	Energies within ≤ 1 kcal/mol per formula unit of QMC or curated experiment; band gaps within 0.2 eV of experiment through a generalized Kohn–Sham scheme whose eigenvalue gap is the fundamental gap [6].
Large and disordered systems: amorphous SiO_2, a metal slab, a solvated protein or an electrochemical interface, 10^4–10^5 atoms	Energies and forces within the calibration tolerances of the $O(N^3)$ reference where that reference is computable, O(N) or O(N log N) measured scaling over at least two decades of system size, and stable energy-conserving molecular dynamics.
Extensions: excitation energies (linear-response and real-time TDDFT [7]), spin–orbit splittings in heavy-element systems (relativistic DFT), superconducting transition temperatures (SCDFT [16])	The analogous accuracy or cost metrics for the chosen generalization against established benchmark sets, with the new functional or algorithm derived from the same exact conditions.

Theory track

Alongside the predictive method, the challenge accepts results on the exact functional itself: tighter bounds or higher-order expansions of known properties (the Lieb–Oxford constant [12], the gradient expansion, scaling relations), and new exact sum rules or constraints that the exact XC functional satisfies and existing approximations violate. A theory

result is scored by proof (formal or expert review), by numerical verification on exact or near-exact densities, and by the improvement obtained on the validation systems when the condition is enforced in a functional.

Source notes

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